VLSI Internship – Task 1

Introduction to VLSI Design Flow and Basic Digital Logic Implementation

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1. What is VLSI?

VLSI (Very Large Scale Integration) is the process of integrating millions to billions of transistors onto a single silicon chip. It forms the foundation of modern electronics such as microprocessors, mobile SoCs, memory chips, GPUs, and AI accelerators. VLSI chips are found in smartphones, computers, automotive electronics, medical devices, and IoT systems.

2. VLSI Design Flow

- 1. System Specification — Define what the chip should do, along with performance, power, and area constraints.
- 2. Architecture Design — Create a high-level block diagram describing functional blocks and data flow.
- 3. RTL Design — Code the design at Register Transfer Level using Verilog / VHDL.
- 4. Functional Verification — Simulate the RTL to confirm the design behaves correctly.
- 5. Synthesis — Convert the verified RTL into a gate-level netlist.
- 6. Physical Design — Perform floorplanning, placement, and routing of the netlist.
- 7. Timing & Power Analysis — Verify the design meets speed and power targets.
- 8. Fabrication — Manufacture the chip on silicon.
- 9. Testing & Validation — Test the fabricated chips to confirm they work as intended.

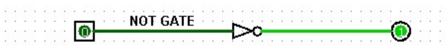
This task (Task-1) focuses on Steps 1 and 2 — System Specification and Architecture Design — and builds the digital logic foundation needed before writing RTL/Verilog code in later tasks.

3. Digital Logic Gates and Truth Tables

Six fundamental logic gates were studied and simulated: AND, OR, NOT, NAND, NOR, and XOR. Their truth tables are shown below.

NOT Gate (Inverts the input)

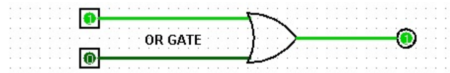
A	Output
0	1
1	0



AND Gate (Output = 1 only if all inputs are 1)

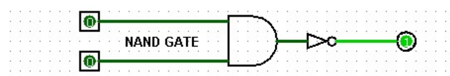
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate (Output = 1 if any input is 1)



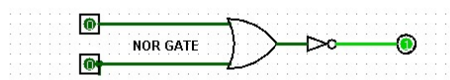
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate (Inverted AND)



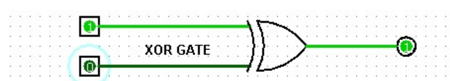
A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate (Inverted OR)



A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate (Output = 1 if inputs are different)



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

4. Combinational Circuit: Half Adder

A Half Adder adds two single-bit binary numbers (A and B) and produces a Sum and a Carry output. It was built using one XOR gate (for Sum) and one AND gate (for Carry).

Sum = A XOR B | Carry = A AND B

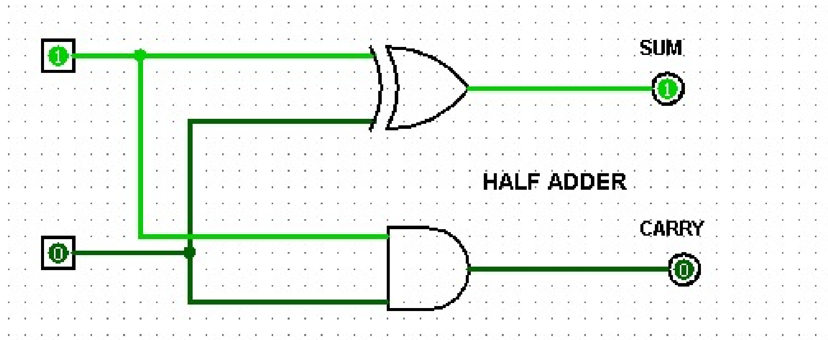


Figure 2: Half Adder circuit simulation showing Sum (XOR) and Carry (AND) outputs.

Half Adder Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

5. Simulation Observations

Inputs used: All four possible combinations of A and B (00, 01, 10, 11) were applied to each gate and to the Half Adder circuit.

Output behavior: Each gate produced outputs matching its expected truth table. The Half Adder correctly produced Sum = 1 only when exactly one input was 1, and Carry = 1 only when both inputs were 1 (i.e., 1+1 = 10 in binary).

Mistakes and corrections: [Fill in based on your own simulation — e.g., initial wiring errors, incorrect gate selection, or input/output pin mismatches, and how they were fixed.]

6. Key Learning Outcomes

- Explained what VLSI is and where it is used
- Described the complete VLSI design flow (System Spec → Testing & Validation)
- Understood digital logic fundamentals and Boolean algebra
- Built and simulated basic logic gates (AND, OR, NOT, NAND, NOR, XOR)
- Designed and verified a Half Adder combinational circuit
- Prepared foundational knowledge for upcoming RTL and Verilog-based tasks